

Stat 5200, Module 3: Completely Randomized Designs and ANOVA

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1 Overview

Objectives and overview

- This material is based Chapters 3–6 on the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Module outcomes include:
 - You will be able to describe a completely randomized design.
 - You will be able to describe the basic model used for ANOVA.
 - You will know how to perform an analysis of data from a CRD using ANOVA.
 - You will be familiar with contrasts and how they are used to perform inference.
 - You will understand the fundamental issues regarding multiple testing, and learn basic methods to overcome these issues.
 - You will know how to assess model assumptions.

2 Completely Randomized Designs

Completely Randomized Design

- Completely randomized design (CRD) is the simplest design for comparing treatments and is sometimes the preferred design.
- *Setup:*
 - There are g treatment groups.
 - N experimental units (EU)s available.
 - You want sample sizes of $n_1, n_2, \dots, n_g$, with $\sum_{i=1}^g n_i = N$.
- *Idea:* All possible sub-group composition are equally as likely.

Performing a Completely Randomized Design

- Randomly choose n_1 EU's for treatment 1.
- Randomly choose n_2 EU's for treatment 2.
- ...
- Randomly choose n_{g-1} EU's for treatment $g - 1$.
- The remaining n_g EU's are assigned to treatment g .
- On a computer, this can be done by permuting $1, 2, \dots, N$, and assigning the first n_1 numbers to treatment 1, the next n_2 to treatment 2, etc.

Advantages and Disadvantages

- Advantages
 - Simple to do; all you need is one random permutation.
 - Simple to explain and understand.
 - Simple to analyze.
 - No restrictions on the sample sizes (other designs may restrict n_i).
 - Easy with missing values.
- Disadvantages
 - Not always possible or cost-effective.
 - The accuracy of the experiment depends on variability among the EUs, not just the variability among treatment groups. Thus, it may require more EUs (\$\$\$\$).

Example: Acid Rain

Wood and Bormann (1974) studied the effect of acid rain on trees. One experiment used 240 six-week-old yellow birch seedlings. These seedlings were divided into five groups of 48 *at random*, and the seedlings within each group received an acid mist treatment 6 hours a week for 17 weeks. The five treatments differed by mist pH: 4.7, 4.0, 3.3, 3.0, and 2.3. Otherwise, the seedlings were treated identically. After the 17 weeks, the seedlings were weighed, and total plant (dry) weight was taken as response.

Questions:

- What type of design is this, and can you describe why?
- What are the EUs, and how many are there?
- What was the response?
- What compromises may have been made in this case?
 - Damage may vary by pH level, plant developmental stage, plant species, etc., yet only pH level was directly addressed.
 - The study wishes to study the impact of acid-rain, but the available treatment was a synthetic acid rain, which may have a slightly different impact.
 - Total plant weight is an important response, but other responses measuring tree health may also be available.

Like many experiments, the researchers necessarily have to narrow down an enormous question to a workable experiment using artificial acid rain under controlled conditions. For any experiment, a considerable amount of non-statistical background work goes into the planning even the simplest experiment.

3 Models and Parameters

Introduction and definitions

- All formal statistical analysis is based on probability models.
- *EX*: The number of heads in ten tosses of a fair coin would be $\text{Binomial}(10, 0.5)$. Here, the model depends on 2 parameters: $N = 10$ the number of trials, $p = 0.5$ is the probability of heads.
- In an experiment, we may posit several different models for the data, all with *unknown* parameters. One of our objectives is often to describe which model is the best description of the data, and make *inference* about parameters.

Definition: Inference

Using data to learn properties of a probability distribution. In our case, using data to estimate parameters of a model.

- The models we will consider for experimental design will have two basic parts:
 - A description of the expected value, or mean, of the data. Sometimes called “model for the means”.
 - A description of how data vary around the treatment means. Sometimes called “model for the errors”.
- For the errors, we will generally assume that they are independent for different data values, have mean zero, and all have a common variance σ^2 .

- To perform formal tests and obtain confidence intervals, we will further assume a distribution of the errors. Generally, we assume that the errors follow a *Gaussian* (Normal) distribution.
- For a CRD, we start with two basic models.
- Recall: g treatment groups, N EUs.

Let y_{ij} be:

- The j th response,
- In the i th treatment group.

Thus, i runs between 1 and g , and j runs between 1 and n_i , in treatment group i . There are many possible models for the mean-treatment effect that we could consider. We will focus on two basic ones:

1. We assume that each treatment has its own mean response μ_i , and that the errors are normally distributed. These assumptions give rise to the *model for the data*:

$$y_{ij} \sim N(\mu_i, \sigma^2).$$

We can alternatively write this same model as:

$$y_{ij} = \mu_i + \epsilon_{ij},$$

where $\epsilon_{ij} \sim N(0, \sigma^2)$ are the independent, normally distributed errors with mean zero and variance σ^2 .

- This is a model with separate group means, which we will sometimes call the *full model*.
2. The next basic model for the means is the *single mean model*, also called the *reduced model*. This model assumes all treatments have the same mean, μ . Combined with the error structure, the single mean model implies that the data are independent and normally distributed with mean μ and constant variance σ^2 :

$$y_{ij} \sim N(\mu, \sigma^2).$$

Again, we may alternatively write this model as:

$$y_{ij} = \mu + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2).$$

- Note that this single mean model is a special case (or restriction) of the group means model: all of the μ_i s equal each other.
- *Model comparison is easiest when one of the models is a restricted or reduced form of the other!*

Defining overall mean

- It's sometimes useful to express the group means μ_i as:

$$\mu_i = \mu^* + \alpha_i.$$

In this case, we call μ^* the *overall mean*, and the α_i 's are the i th *treatment effects*.

The restricted model in this case implies $\alpha_i = 0$ for all $i \in 1 : g$.

Note that if there are g α_i 's, and one overall mean μ^* , this means there are $g + 1$ parameters. This is problematic because they are not uniquely determined. We will avoid this issue by imposing an additional mathematical structure on the set of g group means. Specifically, we either define μ^*

and require that $\alpha_i = \mu_i - \mu^*$, or we define α_i and then set $\mu^* = \mu_i - \alpha_i$. These are really just two-sides of the same coin, and mathematically give us the same information. However, some software will do one or the other, and you should think carefully about what's happening. Ultimately it doesn't matter as long as we can identify which choice is in use at any given time. That is, as long as we can determine the treatment means, the differences of treatment means, and compare the models, it doesn't matter.

- One classic choice is to define μ^* as the mean of the treatment means:

$$\mu^* = \sum_{i=1}^g \mu_i / g,$$

which, defining $\alpha_i = \mu_i - \mu^*$, implies

$$\sum_{i=1}^g \alpha_i = 0.$$

- An alternative choice (that can sometimes make hand-work simpler) instead describes μ^* as the weighted average:

$$\mu^* = \sum_{i=1}^g n_i \mu_i / N,$$

in which case the weighted sum of the treatment effects will be zero:

$$\sum_{i=1}^g n_i \alpha_i = 0.$$

In our current model, defining α_i is a bit of an over-complication, but it will be useful later.

- Another common choice in software is to set $\mu^* = \mu_j$, for some $j \in 1 : g$. Many software just use the last mean: $\mu^* = \mu_g$.

With this choice, we still have our standard restriction:

$$\alpha_i = \mu_i - \mu^*, \quad \text{for all } i \in 1 : g.$$

Thus, we have:

$$\alpha_i = \begin{cases} \mu_j - \mu_j = 0 & \text{if } i = j \\ \mu_i - \mu_j & \text{otherwise} \end{cases}$$

Importantly, it doesn't really matter which is used as long as you know the version being used. You can always re-frame / recover the other form if you prefer it. For instance, the last option is particularly useful if treatment j is some form of control; this way, the treatment effect is always compared to the control.

Notably, the added constraints on α_i s imply that not all are free to vary. With μ^* being fixed (or pre-selected), only $g - 1$ of them are free to vary, the last treatment effect is then determined by whatever is needed to satisfy the constraint. We express this by saying that the treatment effects have $g - 1$ *degrees of freedom*.

Model Recap

However you choose to define μ^* and α_i 's, we end up with the following models for the data:

- *The full model:*

$$y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$$

- *The reduced model:*

$$y_{ij} = \mu + \epsilon_{ij}$$

4 Parameter Estimation

Estimating parameters

- We now want to consider how to compare these two models using data. The classic tool that we will be using is called ANalysis Of VAriance (ANOVA). In practice, we will just be using some form of statistical software (in this class, it will be R). Part of the output of ANOVA will be parameter estimates and other important statistics, so we will first describe what these estimates are and how we express them mathematically.
- We want to use data to estimate μ_i 's and the variance σ^2 . From these, we'll compute μ^* and α_i 's.
- We will focus on computing *unbiased* estimates of these parameters. Though there is a technical definition of *unbiased estimators*, we will simply use this to mean that if we were to average the values of the estimates across all *potential* random errors, we would get the true parameter values. We will use the “hat” notation ($\hat{\bullet}$) throughout the class, to distinguish a true parameter value from its estimate. For instance, $\hat{\mu}$ is an estimator of μ .

- Estimate for μ_i :

$$\hat{\mu}_i = \bar{y}_{i\bullet} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$$

- The estimate for μ^* is fixed, so it depends on how we defined it.

- Unweighted average.

$$\hat{\mu}^* = \frac{1}{g} \sum_{i=1}^{n_i} \hat{\mu}_i = \frac{1}{g} \sum_{i=1}^{n_i} \bar{y}_{i\bullet},$$

$$\hat{\alpha}_j = \hat{\mu}_j - \hat{\mu}^* = \bar{y}_{j\bullet} - \frac{1}{g} \sum_{i=1}^{n_i} \bar{y}_{i\bullet}$$

- Weighted average.

$$\hat{\mu}^* = \frac{1}{N} \sum_{i=1}^g n_i \hat{\mu}_i = \frac{1}{N} \sum_{i=1}^g n_i \bar{y}_{i\bullet} = \frac{1}{N} \sum_{i=1}^g \sum_{j=1}^{n_i} y_{ij} = \bar{y}_{\bullet\bullet}$$

$$\hat{\alpha}_j = \hat{\mu}_j - \hat{\mu}^* = \bar{y}_{j\bullet} - \bar{y}_{\bullet\bullet}$$

- Reference Group.

$$\hat{\mu}^* = \hat{\mu}_j = \bar{y}_{j\bullet}$$

$$\hat{\alpha}_j = \hat{\mu}_j - \hat{\mu}^* = 0$$

$$\hat{\alpha}_i = \hat{\mu}_i - \hat{\mu}^* = \bar{y}_{i\bullet} - \bar{y}_{j\bullet}$$

- The only parameter left to estimate is σ^2 :

$$\hat{\sigma}^2 = \text{MS}_E = \frac{\text{SS}_E}{N - g} = \frac{\sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2}{N - g}$$

We use the notation SS_E to denote the “Sum of Squares for Error”, and MS_E to mean the “Mean Square for Error”. The denominator includes $N - g$ as the number of *degrees of freedom*. Consider the summand: $(y_{ij} - \bar{y}_{i\bullet})^2$. This is the deviations from the group mean, and they must add to zero in any group. Thus, any $n_i - 1$ of them determine the remaining value, so that in each group, there are only $n_i - 1$ degrees of freedom, or that there are $N - g$ degrees of freedom for error for the experiment.

- Above, we have described *point estimates*. In some sense, these are our “best guess” as to the value of a particular parameter.
- We also want some measure of uncertainty of this estimate. The standard frequentist way of doing this is a *confidence interval*.
- Confidence intervals for α are not very useful, because it depends on how we define them. Confidence intervals for group means are typically defined as:

$$\text{unbiased estimate} \pm \text{multiplier} \times (\text{estimated})\text{SE of estimate.}$$

5 Comparing Models: ANOVA

Analysis of Variance

- In our analysis of a CRD, our models focus on the *mean responses* of the treatment groups.
- A basic question we might want answered: *Are all the means the same, or are some different?*
- We can express this same question mathematically:
Is the reduced model (single mean) an adequate description of the data, or do we need the model of separate treatment group means?
- Recall that the single-mean model is just a special case of the separate mean model!
- *Analysis of Variance*, abbreviated ANOVA, is a method for comparing the fit of two models, one a reduced version of the other.
- The basic idea for many statistical tests (including ANOVA) follows *Occam’s Razor*: we use the simplest model that is consistent with the data; we will only use a more complicated model if the data indicate that the more complicated model is needed.
- How do we know if a more complex model is needed? We compare *residuals*.

A *residual* r_{ij} is the difference between the data y_{ij} and the fitted mean-model. Using the weighted average definition for the mean (e.g., $\hat{\mu}^* = \bar{y}_{\bullet\bullet}$):

- For the single-mean model, our best prediction is $\hat{y}_{ij} = \hat{\mu}^* = \bar{y}_{\bullet\bullet}$. Thus, the residual is

$$r_{ij}^{(1)} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{\bullet\bullet}.$$

- For the separate means model, the fitted values are the means of each group: $\hat{y}_{ij} = \hat{\mu}_i = \bar{y}_{i\bullet}$, and the residuals are:

$$r_{ij}^{(2)} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\bullet}$$

- We then compare model “closeness” by looking at the sum of square residuals (SSR):

$$SSR = \sum_{i=1}^g \sum_{j=1}^{n_j} r_{ij}^2.$$

- **Question:** Why square the residuals?
- Recall that there is some uncertainty in our estimates themselves, and it’s not necessarily the only way to estimate these parameters. **Question:** Why these estimates? **Answer:** The estimates presented are *least squares* estimators, meaning that they are picked to minimize the SSR for for specific model choice.

- The lower the SSR, the more of the variance in the data is explained by the model.
- Importantly, the SSR for the reduced (single mean) model can *never* be lower than the separate model, and in practice, is nearly always larger.
- What we need now is a way to consistently decide when the separate means model is sufficiently “better” than the single means model.
- To do so, we will set up a statistical hypothesis test:

We will let the “null model” be that the single mean model is true, vs the alternative hypotheses that the separate means model is true. We first need to express this hypothesis in terms of parameters, rather than models.

Using the $\mu_i = \mu + \alpha_i$ notation for group means, we can express the null hypothesis H_0 mathematically as:

$$H_0 : \alpha_i = 0, \quad \text{for all } i$$

against

$$H_1 : \alpha_i \neq 0, \quad \text{for some } i$$

Important: The alternative isn’t that all α_i s are different, only that there is at least one treatment group with non-zero α_i .

Note that we have taken our original model point of view (checking if one model has less residual variation), and converted to a parameter point of view (checking if there is sufficiently great variation between the observed group means). Really, these are the same thing.

ANOVA mechanics

- ANOVA works by partitioning the total variance in the data into parts that mimic the model.
- For instance, in the separate mean model, the model says the data differ from the grand mean by a random error and a treatment effect:

$$y_{ij} - \mu = \alpha_i + \epsilon_{ij}$$

ANOVA will have a similar decomposition: one part dealing with group means, one part with deviations from those means:

$$y_{ij} - \bar{y}_{\bullet\bullet} = (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) + (y_{ij} - \bar{y}_{i\bullet}) \quad (1)$$

$$= \hat{\alpha}_i + r_{ij} \quad (2)$$

On the left, we have the deviation of a single response from the grand mean. If we square them, then add them up, we get the **Total Sum of Squares** (SS_T):

$$SS_T = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2.$$

The first difference on the right is the estimated treatment effect, $\hat{\alpha}_i$. If we square the treatment effects, and add them up, we get the **Treatment Sum of Squares** (SS_{Trt}):

$$\begin{aligned} SS_{Trt} &= \sum_{i=1}^g \sum_{j=1}^{n_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 \\ &= \sum_{i=1}^g n_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 \\ &= \sum_{i=1}^g n_i \hat{\alpha}_i^2 \end{aligned}$$

The last expression can be readily interpreted as: square each treatment effect, multiply by the number of units receiving that effect, and add over the levels of the effect.

The second difference on the right of Eqs. 1–2 is the ij th residual from the model, r_{ij} . If you squared and added these, we get the **Error Sum of Squares** (SS_E):

$$SS_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2$$

Note that this is the same SS_E that we used to estimate σ^2 .

- Recall from Eq. 2, we have:

$$y_{ij} - \bar{y}_{\bullet\bullet} = \hat{\alpha}_i + r_{ij},$$

so squaring both sides gives:

$$(y_{ij} - \bar{y}_{\bullet\bullet})^2 = \hat{\alpha}_i^2 + r_{ij}^2 + 2\hat{\alpha}_i r_{ij}$$

- Adding over i and j :

$$SS_T = SS_{Trt} + SS_E + 2 \sum_{i=1}^g \sum_{j=1}^{n_i} \hat{\alpha}_i r_{ij}$$

It turns out that the last sum is equal to zero (6200-HW problem). This means:

$$SS_T = SS_{Trt} + SS_E$$

- Finally, we can connect this back to the residuals.
- The Sum of Squared Residuals (SSR) depended on our model:

$$SSR_{\text{single means}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2 = SS_T$$

$$SSR_{\text{separate means}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2 = SS_E$$

Thus,

$$SS_T = SS_{Trt} + SS_E \implies SS_{Trt} = SSR_{\text{single means}} - SSR_{\text{separate means}}$$

We have found a link between testing equality of group means and comparing models via SSR of two models: If we fix a dataset, then SS_T is fixed. If there is a lot of variation *between* groups (i.e., large SS_{Trt}), then that must mean there is less variation from the treatment means (i.e., smaller SS_E).

Calibrating Effect Size

- We have established:
 - We will compare single mean against separate mean by comparing their SSRs.
 - Comparing SSR can be done by decomposing total error into two parts: $SS_T = SS_{T_{rt}} + SS_E$, and if the separate model is “much better” than the $SS_{T_{rt}}$ will be large.
 - **How large is large enough?** Part of the problem is that $SS_{T_{rt}}$ is data specific. For instance, if our response being measured is some type of length, the size of $SS_{T_{rt}}$ will change if we chose to use inches or centimeters. Even for the same units, what constitutes a large difference depends heavily on the problem being asked! Consider drawing two diagrams, both with the same treatment means. Even though $SS_{T_{rt}}$ may be the same in each diagram, how confident we are depends on the spread of data around each mean (SS_E). If the spread is small, we should readily detect differences in treatment means, whereas if the spread is large it becomes difficult to tell if the means are truly different.
 - The answer also lies in the decomposition of sum of squares.
- Following the single-mean model (H_0),

$$y_{ij} \sim N(\mu, \sigma^2).$$

This implies that

$$\frac{SS_E}{\sigma^2} = \sum_{i=1}^g \sum_{j=1}^{n_i} \frac{(y_{ij} - \bar{y}_{i\bullet})^2}{\sigma^2} \sim \chi_{N-g}^2,$$

and

$$\frac{SS_{T_{rt}}}{\sigma^2} = \sum_{i=1}^g \sum_{j=1}^{n_i} \frac{(\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2}{\sigma^2} \sim \chi_{g-1}^2,$$

and therefore

$$\begin{aligned} F &= \frac{(\frac{SS_{T_{rt}}}{\sigma^2})/(g-1)}{(\frac{SS_E}{\sigma^2})/(N-g)} \\ &= \frac{SS_{T_{rt}}/(g-1)}{SS_E/(N-g)} \\ &= \frac{MS_{T_{rt}}}{MS_E} \sim F_{g-1, N-g} \end{aligned}$$

Where $MS_{T_{rt}}$ and MS_E correspond to the mean-square treatment and mean-square error, respectively, using the degrees of freedom of each as the denominator. For this to hold, the numerator and denominator need to be independent random variables; fortunately for us, our null model (H_0) implies this is the case.

ANOVA variable recap

- Sum of Square Residuals:

$$SSR = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \hat{y}_{ij})^2$$

- Treatment Sum of Squares:

$$SS_{T_{rt}} = \sum_{i=1}^g \sum_{j=1}^{n_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 = \sum_{i=1}^g n_i \hat{\alpha}_i^2$$

- Error Sum of Squares:

$$SS_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2$$

- Total Sum of Squares:

$$SS_T = SS_{Trt} + SS_E = \sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2$$

- F-statistic:

$$\frac{SS_{Trt}/(g-1)}{SS_E/(N-g)} \sim F_{g-1, N-g}$$

Example in R

We are going to look at an example using the `PlantGrowth` dataset in R. This is a built-in dataset, so we don't need to download anything.

```
library(emmeans) # Package that we will use for analysis
data(PlantGrowth)
str(PlantGrowth)

'data.frame': 30 obs. of 2 variables:
 $ weight: num  4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.5..
 $ group : Factor w/ 3 levels "ctrl","trt1",...: 1 1 1..

levels(PlantGrowth$group)

[1] "ctrl" "trt1" "trt2"
```

```
pg_model <- aov(weight ~ group, data = PlantGrowth)
summary(pg_model)

          Df Sum Sq Mean Sq F value Pr(>F)
group      2  3.766   1.8832   4.846 0.0159 *
Residuals 27 10.492   0.3886
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The first row of the table gives the degree of freedom, sum of squares, and mean square for the treatment, as well as the F statistic and corresponding P-value. The bottom row gives the same values for the residuals (or errors), giving the DF, SS_E and MS_E .

Now to get group means:

```
pg_emm <- emmeans(pg_model, ~ group)
pg_emm

group emmean    SE df lower.CL upper.CL
ctrl    5.03 0.197 27     4.63     5.44
trt1    4.66 0.197 27     4.26     5.07
trt2    5.53 0.197 27     5.12     5.93

Confidence level used: 0.95
```

6 Contrasts

Introduction to Contrasts

- An ANOVA can give us a good overview as to whether all the treatment groups have the same mean response.
- However, an ANOVA by itself does not tell which treatments are different or in what ways they differ.
 - One of the problems: There are many ways to define treatment effects, so confidence intervals are not immediately useful.
- One method to examine treatment effects is called a *contrast*. The initial idea of a contrast is quite simple, and intuitive: we want a formula which compares two or more treatment means or effects in a way that does not depend on the overall level μ .

For example, consider an experiment with one control group, two experimental treatment groups. If we want to estimate the different in group means from the treatment (μ_1) and only one of the experiments (μ_2), we may consider computing:

$$\mu_1 - \mu_2 = (\mu^* + \alpha_1) - (\mu^* + \alpha_2) = \alpha_1 - \alpha_2$$

We will use contrasts to generalize this idea and to build useful ways of comparing not only distinct groups, but combinations of groups.

Formal Definition

Definition: Contrasts

A contrast is a linear combination of treatment means, μ_i 's:

$$W(\mu_{1:g}) = \sum_i w_i \mu_i, \quad \text{with} \quad \sum_i w_i = 0$$

- The condition $\sum_i w_i = 0$ guarantees that $W(\mu_{1:g})$ does not depend on μ .

$$\begin{aligned} W(\mu_{1:g}) &= \sum_{i=1}^g w_i \mu_i \\ &= \sum_{i=1}^g w_i (\mu^* + \alpha_i) \\ &= \sum_i (w_i \mu^*) + \sum_i (w_i \alpha_i) \\ &= \sum_i (w_i \alpha_i) \end{aligned}$$

- Often the weights w_i themselves are referred to as a contrast.
- An *observed* contrast or estimated contrast is defined as:

$$W(\bar{y}_{1:g,\bullet}) = \sum_i w_i \bar{y}_{i\bullet} = \sum_i w_i \hat{\alpha}_i = W(\hat{\alpha}_{1:g})$$

6.1 Common Contrasts

Types of Common Contrasts

Pairwise Contrasts

- It is often of interest to compare all pairs of groups.
- For a total of g treatment groups, there are $\frac{1}{2}g(g-1)$ different pairwise contrasts.

For example, suppose there are 4 treatment groups. Then there are $\binom{4}{2} = 6$ pairwise contrasts that make up a *family* of contrasts, with the following vectors of w_i 's:

$$(1, -1, 0, 0), (1, 0, -1, 0), (1, 0, 0, -1), (0, 1, -1, 0), (0, 1, 0, -1), (0, 0, 1, -1)$$

Comparison with a Control

- Let's suppose treatment 1 is a control.
- An obvious idea is to compare the mean or effect of the control with the average mean or effect of all non-controls.

For example, consider one control, $g-1$ alternative treatments, then, in addition to pairwise comparisons with the control, one contrast might be comparing the mean of the control to the mean of all non-controls:

$$\mu_1 - (\mu_2 + \mu_3 + \dots + \mu_g)/(g-1)$$

Note that it doesn't matter how we scale the contrasts, as long as the coefficients sum to zero.

Multiple Treatments (Factorial Designs)

- Suppose there are two treatment factors, A and B, each at two levels, 1 and 2.
- We'll denote the treatment group means as $\mu_{A1}, \mu_{A2}, \mu_{B1}, \mu_{B2}$, respectively.
- We might want to compare the averages of just one treatment type (A vs B or 1 vs 2).

In this scenario, contrasts of interest may include:

$$\frac{1}{2}(\mu_{A1} + \mu_{A2}) - \frac{1}{2}(\mu_{B1} + \mu_{B2})$$

or

$$\frac{1}{2}(\mu_{A1} + \mu_{B1}) - \frac{1}{2}(\mu_{A2} + \mu_{B2})$$

This way, we can measure the effect of one treatment, while "averaging out" the effect of the other.

- What do we do with contrasts?
- If a contrast is large (in magnitude), that provides evidence that there is a difference between treatment levels that are specified by the contrast. Conversely, if it is small, there is little evidence of a difference.
- The common question arises: how large is "large enough" to declare the contrast is not zero?
- We can use the linearity of the model to answer this question!

6.2 Variance and Confidence Intervals

Inference on Contrasts

- Under the null hypothesis and normality assumption, we can use Student's t with $df = N - g$ in tests or confidence intervals.
- Under the null hypothesis, $W(\mu_i) = \sum_i w_i \mu_i = 0$.
- The standardized estimated contrast follows a student-t distribution with $df = N - g$:

$$t = \frac{\sum_i w_i \bar{y}_{i\bullet}}{\sqrt{MS_E} \sqrt{\sum_i w_i^2 / n_i}}, \quad MS_E = \frac{\sum_{i=1}^g \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\bullet})^2}{N - g}$$

- A $1 - \alpha$ confidence interval for a contrast $W(\mu_i)$ has the usual form:

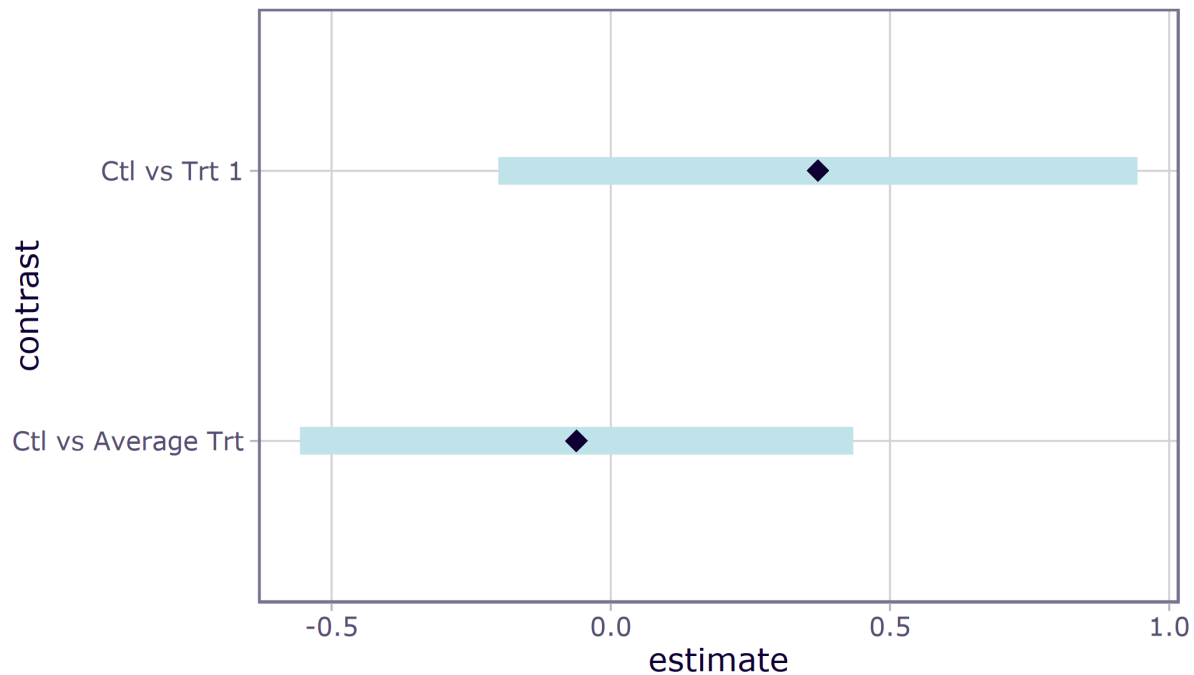
estimate $\pm$ critical value $\times$ standard error

Contrasts in R

```
# pg_model <- aov(weight ~ group, data = PlantGrowth)
# pg_emm <- emmeans(pg_model, ~ group)
custom_list <- list(
  "Ctl vs Average Trt" = c(1, -0.5, -0.5),
  "Ctl vs Trt 1" = c(1, -1, 0)
)
contrast(pg_emm, method = custom_list)
```

contrast	estimate	SE	df	t.ratio	p.value
Ctl vs Average Trt	-0.0615	0.241	27	-0.255	0.8009
Ctl vs Trt 1	0.3710	0.279	27	1.331	0.1944

```
pg_cont <- contrast(pg_emm, method = custom_list)
plot(pg_cont) # level option to change 1-alpha
```



7 Multiple Comparisons

Basic Error Types

- A test of a single hypothesis is characterized by two error rates: the Type I error rate (α) and the Type II error rate (β).

The Need for Multiple Comparisons

- When picking contrasts, it is important to consider the issue of *multiple comparisons*.
- Idea: we pick a α value to test at (e.g., $\alpha = 0.05$). That means, for a single test, there is only a 5% chance (1/20) that we will “incorrectly reject” a true null. **What if we conduct 15 tests?**
If you conduct $K = 15$ independent tests at $\alpha = 0.05$, the probability of making *at least one* Type I error is:

$$1 - (1 - 0.05)^{15} \approx 0.54$$

There is a 54% chance of a false positive result across the study!

To maintain scientific rigor, we must adjust our significance thresholds or p -values based on the total number and nature of the comparisons.

Types of Error Rates

When conducting K hypotheses, the situation becomes more complicated, and we must define which type of error rate we to control.

- *Per Comparison Error Rate (PCER)*: The probability of a Type I error for a single, specific hypothesis test (standard α).
- *Familywise Error Rate (FWER)*: The probability of making at least one Type I error among K tests, assuming *all* null hypotheses are true.
- *Strong Familywise Error Rate (Strong FWER)*: The probability of making at least one Type I error among true nulls, *regardless* of how many null hypotheses are true or false.
- *False Discovery Rate (FDR)*: The expected proportion of false discoveries (false rejections) among all rejected null hypotheses:

$$\text{FDR} = E \left[\frac{\text{False Rejections}}{\text{Total Rejections}} \right]$$

Controlling Strong FWER guarantees that even if 9 out of 10 treatments are truly effective, the probability of falsely declaring the 10th ineffective treatment as effective remains bounded by α .

Controlling FDR allows for some false positives in exchange for substantially higher statistical power, making it ideal for high-throughput screening where follow-up validation will occur.

7.1 General Multiple Testing Methods approaches

General Methods: Introduction

- We will start by introducing some general methods that can be applied *regardless* of the model assumptions or type of hypothesis being adjusted.
- The two most general are the *Bonferroni* and *Holm* adjustment methods, that require no assumptions.
- These can be easily implemented using the `p.adjust` method in R:

```
p.adjust(
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),
  method = 'holm'
)

[1] 0.136 0.006 0.168 0.280 0.168 0.100
```

General Methods: Bonferroni Adjustment

- **Description:** A simple, conservative adjustment based on the Bonferroni inequality ($P(\bigcup E_i) \leq \sum P(E_i)$). Makes no distributional assumptions and works for any set of K tests.
- **Error Controlled:** Controls the *Strong Familywise Error Rate (Strong FWER)* at level α .
- **How-To Implement:**
 - *Adjust Significance Level:* Reject $H_{0,i}$ if $p_i \leq \alpha_{\text{adj}} = \frac{\alpha}{K}$.
 - *Adjust p-values:* Multiply raw p -values by K :

$$p_{i,\text{adj}} = \min(1, K \times p_i)$$

- **When to Use:** Best for a small, pre-planned set of $K \leq 6$ contrasts. Becomes overly conservative as K increases.

```
p.adjust(
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),
  method = 'bonferroni'
)

[1] 0.204 0.006 0.336 1.000 0.360 0.120
```

General Methods: Holm's Step-Down Method

- **Description:** A sequential step-down modification of Bonferroni. It is strictly and uniformly more powerful than standard Bonferroni while making zero distributional or independence assumptions.
- **Error Controlled:** Controls the *Strong Familywise Error Rate (Strong FWER)* at level α .
- **How-To Implement::**
 1. Sort raw p -values from smallest to largest: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(K)}$.
 2. Compare each $p_{(i)}$ sequentially to the adaptive threshold:

$$\text{Threshold}_i = \frac{\alpha}{K - i + 1}$$

3. Start at $i = 1$. As long as $p_{(i)} \leq \frac{\alpha}{K - i + 1}$, reject $H_{(i)}$ and proceed to $i + 1$.
 4. **Stop Rule:** The moment $p_{(i)} > \frac{\alpha}{K - i + 1}$, *stop testing* and retain all remaining null hypotheses $H_{(i)}, \dots, H_{(K)}$.
- Alternatively, we can adjust the p -values using the formula:

$$p_{\text{adj},i} = \max_{j \leq i} \left\{ \min \left\{ (K - j + 1)p_{(j)}, 1 \right\} \right\}$$

For other methods that control the Strong FWER, there is the Hochberg and Hommel methods, but they require that the hypothesis tests are independent. See the orthogonal contrasts section of chapter 5 of Oehlert (2000) for some possible examples.

```
p.adjust(
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),
  method = 'holm'
)

[1] 0.136 0.006 0.168 0.280 0.168 0.100
```

General Methods: Benjamini-Hochberg (FDR)

- **Description:** A sequential step-up procedure designed to maximize power when testing hundreds or thousands of hypotheses (e.g., genomics, large screening studies).
- **Error Controlled:** Controls the *False Discovery Rate (FDR)* at target level q (e.g., $q = 0.05$).
- **How-To Implement:**
 1. Sort raw p -values from smallest to largest: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(K)}$.
 2. Calculate the FDR critical boundary for each rank i :

$$\text{Critical Value}_i = \left(\frac{i}{K} \right) q$$

3. Find the *largest* rank k such that $p_{(k)} \leq \left(\frac{k}{K} \right) q$.
4. **Decision Rule:** Reject $H_{(1)}, H_{(2)}, \dots, H_{(k)}$. (All hypotheses ranked up to k are declared significant, even if individual $p_{(i)} > \text{Critical Value}_i$ for $i < k$).

```
p.adjust(
  c(0.034, 0.001, 0.056, 0.28, 0.06, 0.02),
  method = 'BH' # or 'fdr', both give the same result
)

[1] 0.068 0.006 0.072 0.280 0.072 0.060
```

7.2 ANOVA and Contrast-Specific Methods

ANOVA specific methods

- If we want to use multiple-testing approaches for ANOVA and contrasts, any of the above are perfectly valid.
- However, we can often use our model assumptions to get more powerful methods specific to ANOVA and our linear models.
- We will next describe a few key methods in this scenario.

ANOVA Methods: Tukey's HSD (All Pairwise)

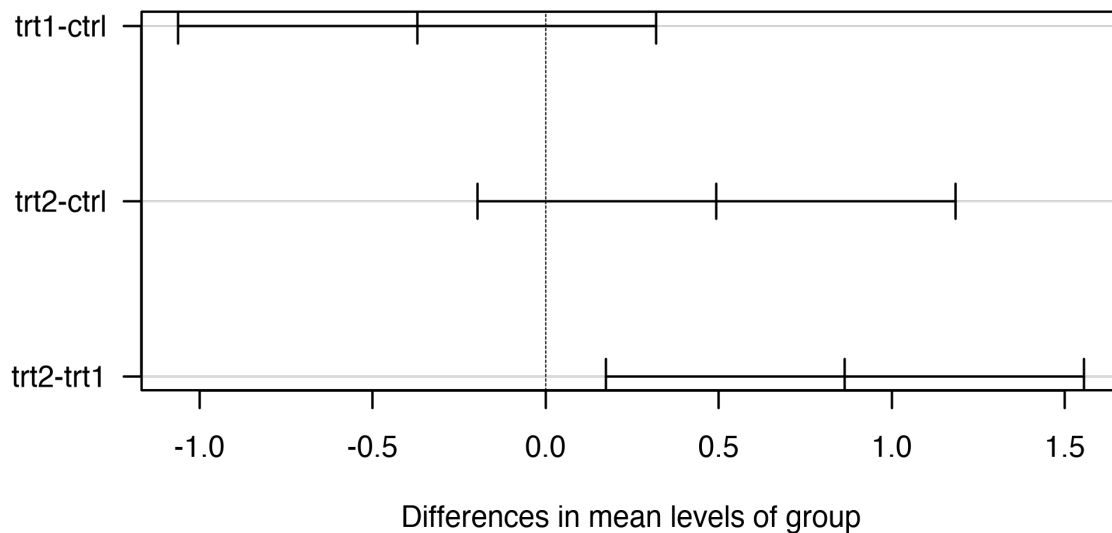
- **Description:** Tailored specifically for comparing all possible pairwise means ($\binom{g}{2}$ comparisons). Uses the Studentized Range distribution (Q) rather than the standard t -distribution.
- **Error Controlled:** Controls the *Strong Familywise Error Rate (Strong FWER)* at level α specifically across the family of all pairwise comparisons.
- **How-To Implement:**
 - Compute the Honest Significant Difference (HSD) threshold:

$$\text{HSD} = q_{1-\alpha}(g, N - g) \sqrt{\frac{MS_E}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

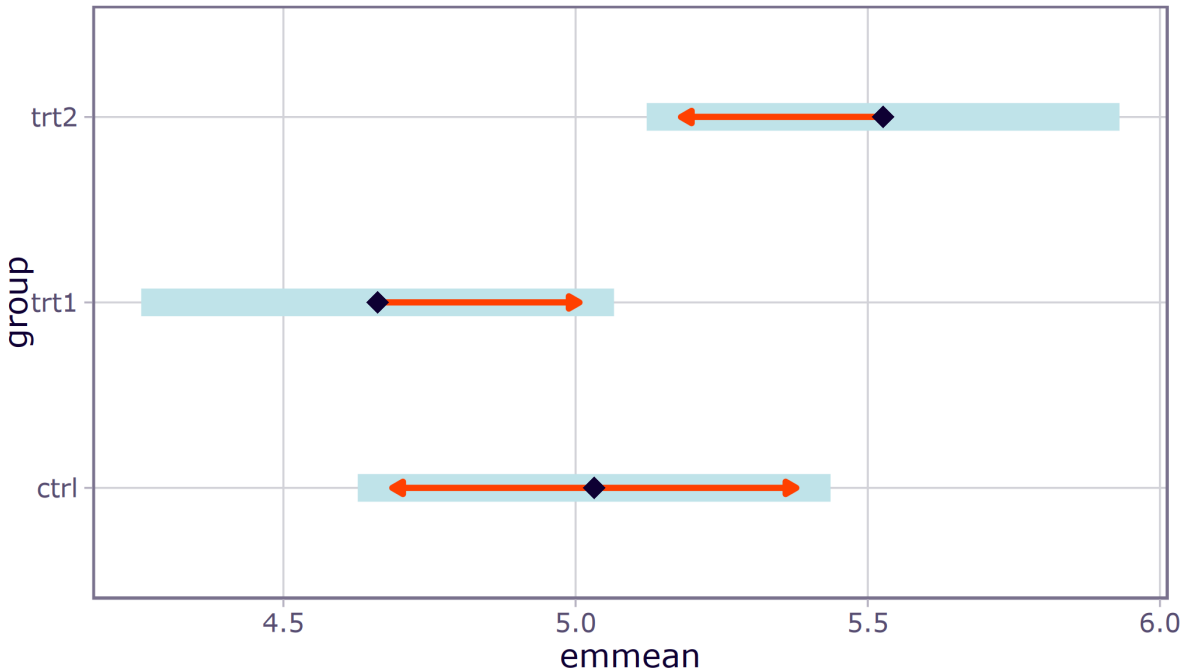
- Declare two group means significantly different if $|\bar{y}_{i\bullet} - \bar{y}_{j\bullet}| \geq \text{HSD}$.
- **Unequal Sample Sizes:** Modern software automatically incorporates the *Tukey-Cramer* modification above, adjusting SE for unequal n_i and n_j .

```
# pg_model <- aov(weight ~ group, data = PlantGrowth)
# pg_emm <- emmeans(pg_model, ~ group)
tukeyRes <- TukeyHSD(pg_model) # Compute Tukey Adjusted Intervals
plot(tukeyRes, las = 1) # Plot Tukey Adjusted Intervals
```

95% family-wise confidence level



```
# Default adjust = 'tukey'.
# Group means differ significantly if arrows don't overlap.
plot(pg_emm, comparisons = TRUE)
```



ANOVA Methods: Dunnett's Test (Control vs Treatments)

- **Description:** Designed specifically for experiments with one baseline control group and $g - 1$ treatment groups. Uses a multivariate t -distribution that accounts for the shared correlation between comparisons involving the same control group.
- **Error Controlled:** Controls the *Strong Familywise Error Rate (Strong FWER)* at level α across the $g - 1$ treatment-vs-control comparisons.
- **How-To Implement:**

- Compute the Dunnett threshold for group i vs control group c :

$$D = d_{1-\alpha}(g - 1, N - g) \sqrt{MS_E \left(\frac{1}{n_i} + \frac{1}{n_c} \right)}$$

- Reject $H_0 : \mu_i = \mu_c$ if $|\bar{y}_{i\bullet} - \bar{y}_{c\bullet}| \geq D$.

- **Advantage:** Substantially more powerful than Tukey or Bonferroni when you only care about comparisons against a control group.

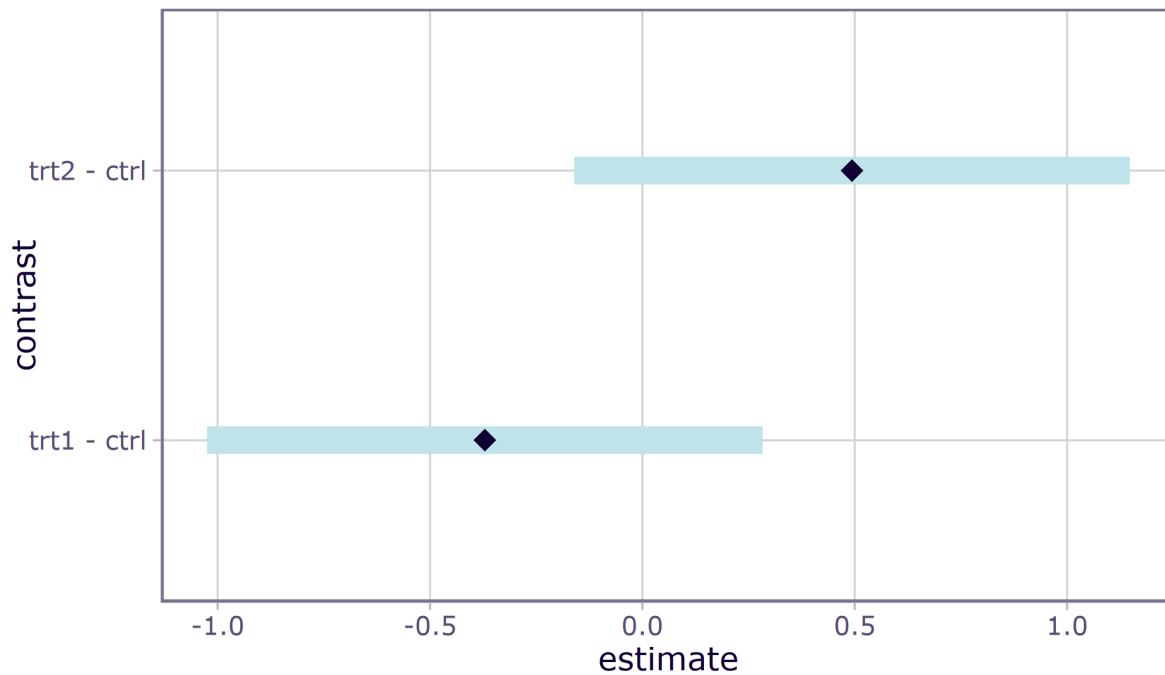
```
# pg_model <- aov(weight ~ group, data = PlantGrowth)
# pg_emm <- emmeans(pg_model, ~ group)

# Compute Dunnett-adjusted contrasts against baseline 'ctrl'
dunnettRes <- contrast(
  pg_emm,
  method = "trt.vs.ctrl",
  ref = "ctrl"
)
summary(dunnettRes)
```

```
contrast    estimate    SE df t.ratio p.value
trt1 - ctrl  -0.371 0.279 27  -1.331  0.3296
trt2 - ctrl   0.494 0.279 27   1.772  0.1582
```

P value adjustment: dunnett method for 2 tests

```
# Plot Dunnett-adjusted contrast confidence intervals
plot(dunnettRes)
```



ANOVA Methods: Scheffé's Method (Data Snooping)

- **Description:** Based on the omnibus ANOVA F -distribution. It is the most conservative post-hoc method, protecting against false positives across *all possible* linear contrasts.
- **Error Controlled:** Controls the *Strong Familywise Error Rate (Strong FWER)* at level α across the *infinite set* of all possible linear contrasts.

- **How-To Implement:**

- For any contrast $\hat{W} = \sum w_i \bar{y}_{i\bullet}$, compute the Scheffé Critical Value (SSD):

$$\text{SSD} = \sqrt{(g-1) F_{1-\alpha, g-1, N-g}} \times \text{SE}(\hat{W})$$

$$\text{where } \text{SE}(\hat{W}) = \sqrt{MS_E \sum \frac{w_i^2}{n_i}}.$$

- Reject $H_0 : W = 0$ if $|\hat{W}| \geq \text{SSD}$.

- **When to Use:** Use when testing custom contrasts constructed *after* inspecting the data ("data snooping") or testing a large number of non-pairwise complex contrasts.

```

# Define custom complex contrasts
# Example: Control vs Average of Treatments, and Trt1 vs Trt2
custom_contrasts <- list(
  "Ctrl vs Avg(Trt1, Trt2)" = c(1, -0.5, -0.5),
  "Trt1 vs Trt2"           = c(0, 1, -1)
)

# Compute Scheffe-adjusted contrasts
scheffeRes <- contrast(pg_emm, method = custom_contrasts,
  adjust = "scheffe")
summary(scheffeRes)

contrast      estimate    SE df t.ratio
Ctrl vs Avg(Trt1, Trt2) -0.0615 0.241 27  -0.255
Trt1 vs Trt2          -0.8650 0.279 27  -3.103
p.value
0.9681
0.0163

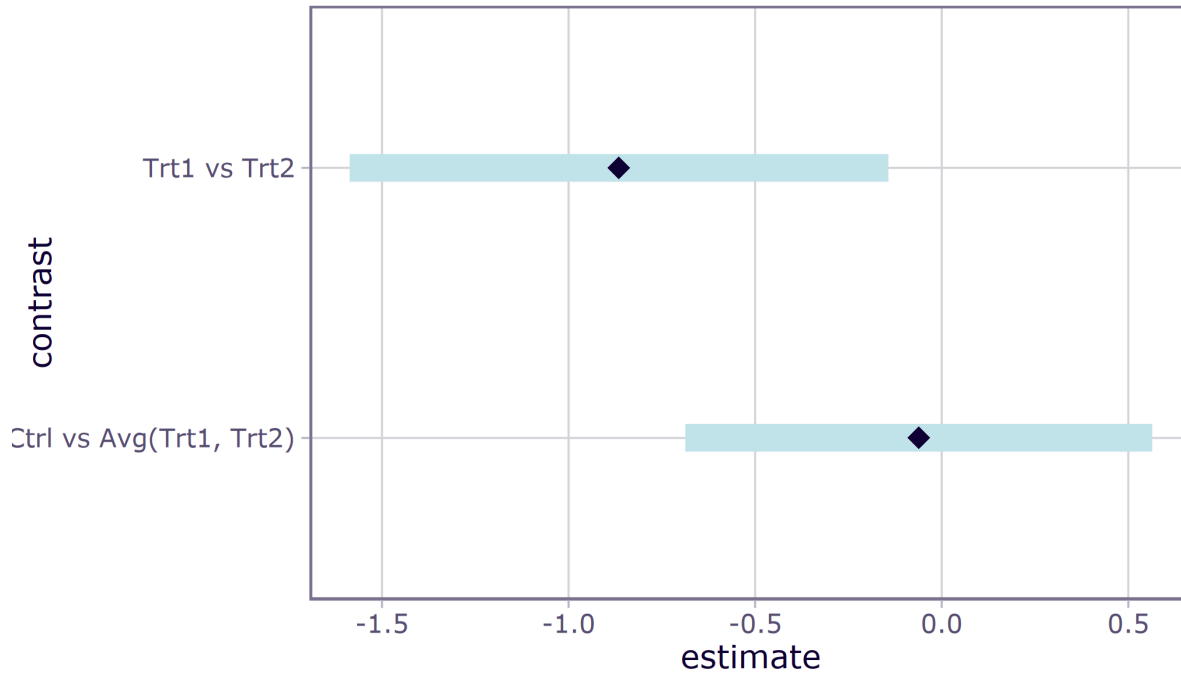
P value adjustment: scheffe method with rank 2

```

```

# Plot Scheffe-adjusted contrast confidence intervals
plot(scheffeRes)

```



Simultaneous Confidence Intervals

- **Definition:** A set of intervals $[L_1, U_1], \dots, [L_K, U_K]$ are $(1 - \alpha)$ simultaneous confidence intervals if the probability that *all* K intervals cover their true parameter values $\theta_1, \dots, \theta_K$ simultaneously is at least $1 - \alpha$:

$$P\left(\bigcap_{k=1}^K \{\theta_k \in [L_k, U_k]\}\right) \geq 1 - \alpha$$

- **General Structure:** For linear contrasts $\theta_k = \sum w_i \mu_i$, simultaneous intervals take the standard form:

$$\hat{\theta}_k \pm M_\alpha \times \text{SE}(\hat{\theta}_k)$$

where M_α is an adjusted critical multiplier derived from the multiple testing procedure:

- **Bonferroni:** $M_\alpha = t_{1-\frac{\alpha}{2K}, N-g}$
- **Tukey’s HSD:** $M_\alpha = \frac{1}{\sqrt{2}} q_{1-\alpha, g, N-g}$
- **Scheffé:** $M_\alpha = \sqrt{(g-1) F_{1-\alpha, g-1, N-g}}$

Duality: Simultaneous CIs and Strong FWER

- **The Duality Principle:** Hypothesis tests and confidence intervals are inverted versions of the same mathematical statement:

$$\text{Reject } H_{0,k} : \theta_k = 0 \text{ at level } \alpha \iff 0 \notin \text{CI}_k^{(1-\alpha)}$$

- **Why Strong FWER is Required:**

- *Weak FWER* only guarantees protection when *all* null hypotheses are true ($\theta_1 = \theta_2 = \dots = \theta_K = 0$).
- In practice, some contrasts will be zero and others will be non-zero. Confidence intervals must cover the true parameters *regardless of which or how many parameters are non-zero*.
- *Strong FWER* controls the Type I error rate under *any* combination of true and false null hypotheses. Inverting a test procedure that controls Strong FWER guarantees $(1 - \alpha)$ simultaneous coverage across all parameters.

- **Practical Takeaway:** When presenting a plot or table of Tukey, Bonferroni, or Scheffé simultaneous confidence intervals, any interval that excludes 0 corresponds *exactly* to a contrast that is statistically significant under that method’s Strong FWER-adjusted test at level α .

Pedagogical Note on Weak vs. Strong FWER for Intervals: To see why Strong FWER is essential for simultaneous confidence intervals, imagine an experiment with $K = 4$ contrasts, where 2 treatment effects are truly zero and 2 are non-zero. A method that only controls Weak FWER (like an unadjusted ANOVA F-test followed by Fisher’s LSD) assumes all 4 effects are zero. Because that assumption is false in this scenario, Weak FWER offers no guarantee that the intervals will cover the true non-zero parameter values 95% of the time.

Strong FWER procedures (such as Tukey or Bonferroni) guarantee that the probability of making *any* error—falsely rejecting a zero contrast OR failing to cover a true parameter value—remains bounded by $\alpha = 0.05$, no matter what the true underlying parameter values are.

Connecting Theory to R: Extracting M_α and CIs

- Modern R packages like `emmeans` apply the critical multiplier M_α automatically when you request confidence intervals.
- **Duality in Output:** Running `summary()` displays test statistics (p -values), while `confint()` displays simultaneous confidence intervals derived using the exact same M_α .

```

library(emmeans)
# 1. Fit model and get pairwise contrasts
pg_model <- aov(weight ~ group, data = PlantGrowth)
pg_emm <- emmeans(pg_model, ~ group)
pg_pairs <- pairs(pg_emm, adjust = "tukey")

# 2. View Hypothesis Testing (p-values)
summary(pg_pairs)

contrast      estimate      SE df t.ratio p.value
ctrl - trt1    0.371 0.279 27   1.331  0.3909
ctrl - trt2   -0.494 0.279 27  -1.772  0.1980
trt1 - trt2   -0.865 0.279 27  -3.103  0.0120

P value adjustment: tukey method for comparing a family of 3 estimates

# 3. View Simultaneous Confidence Intervals (Estimate +/- M_alpha * SE)
# Uses Tukey multiplier M_alpha = q_{1-alpha} / sqrt(2)
confint(pg_pairs)

contrast      estimate      SE df lower.CL upper.CL
ctrl - trt1    0.371 0.279 27   -0.32   1.062
ctrl - trt2   -0.494 0.279 27   -1.19   0.197
trt1 - trt2   -0.865 0.279 27   -1.56  -0.174

Confidence level used: 0.95
Conf-level adjustment: tukey method for comparing a family of 3 estimates

```

- **Visual Verification:** Observe that for `trt1 - trt2`, the Tukey-adjusted p -value is $0.012 < 0.05$, and its 95% simultaneous CI $[-1.66, -0.15]$ completely excludes 0.

7.3 Summary and Practical Selection

Summary: Selecting an Adjustment Method

- No single adjustment method is universally “best”—the choice depends entirely on the structure of your research question and the family of tests.

8 Applied Example: Plant Growth

Example 1: Data Setup and the F-Test

- We will use R’s built-in `PlantGrowth` dataset, measuring dried plant yields under a control and two different treatment conditions.
- As discussed, a Completely Randomized Design (CRD) is the simplest design for comparing treatments. We have $g = 3$ groups and $N = 30$ total plants.

```

# Load data and check the structure
data(PlantGrowth)
str(PlantGrowth)

```

Method	Target Error Rate	Family Scope	Best Application
Bonferroni	Strong FWER	General (K tests)	Small set of pre-planned contrasts ($K \leq 6$)
Holm	Strong FWER	General (K tests)	Step-down alternative to Bonferroni (more power)
BH / FDR	FDR	General (K tests)	High-throughput screening / large K
Tukey / HSD	Strong FWER	All Pairwise ($\binom{g}{2}$)	Comparing all treatment group pairs
Dunnett	Strong FWER	Control vs All ($g - 1$)	Comparing all treatments against one control
Scheffé	Strong FWER	Infinite Contrasts	Data snooping / Complex post-hoc contrasts

```
'data.frame': 30 obs. of 2 variables:
 $ weight: num 4.17 5.58 5.18 6.11 4.5 4.61 5.17 4.5..
 $ group : Factor w/ 3 levels "ctrl","trt1",...: 1 1 1..

# Ensure the treatment is a factor (categorical)
class(PlantGrowth$group)

[1] "factor"
```

- We first test the null hypothesis that all group means are equal using an omnibus ANOVA F-test.

```
# Run the ANOVA model
pg_model <- aov(weight ~ group, data = PlantGrowth)

# View the ANOVA table
summary(pg_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	3.766	1.8832	4.846	0.0159 *
Residuals	27	10.492	0.3886		

```
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- **Interpretation:** With a p -value of 0.0159, we reject the null hypothesis. There is a significant difference in plant yields between at least some of the groups. But which ones?

Example 2: Formulating Custom Contrasts

- We know from our model that the single mean model is inadequate, so we want to estimate the separate group means.
- Let's test a specific contrast: Is the control group significantly different from the average of the two experimental treatments?

```
library(emmeans)

# Get estimated marginal means
pg_emm <- emmeans(pg_model, ~ group)

# Define contrast: Control (1) vs Average of Trt1 (2) and Trt2 (3)
# Weights must sum to zero
custom_list <- list("Control vs Average Trt" = c(1, -0.5, -0.5))

# Test the contrast
contrast(pg_emm, method = custom_list)

contrast      estimate    SE df t.ratio
Control vs Average Trt -0.0615 0.241 27  -0.255
p.value
0.8009
```

- **Interpretation:** The p -value is 0.2644. We do not have sufficient evidence to say the control is different from the *average* of the two treatments.

Example 3: Multiple Comparisons

- What if we want to compare all individual pairs of treatments?
- We must adjust for multiple comparisons to control the familywise error rate. Tukey's HSD is the preferred method for all pairwise comparisons.

```
# Run all pairwise comparisons with Tukey adjustment
pg_pairwise <- contrast(pg_emm, method = "pairwise")

summary(pg_pairwise)

contrast      estimate    SE df t.ratio p.value
ctrl - trt1      0.371 0.279 27   1.331  0.3909
ctrl - trt2     -0.494 0.279 27  -1.772  0.1980
trt1 - trt2     -0.865 0.279 27  -3.103  0.0120

P value adjustment: tukey method for comparing a family of 3 estimates
```

- **Interpretation:** After applying the Tukey penalty, only the difference between `trt1` and `trt2` is statistically significant ($p = 0.0120$). The control is not significantly different from either individual treatment.

Other Common Scenarios (See Textbook)

- While Holm and Tukey methods cover many of basic the basic contrasts we may wish to test, there are specialized methods for other common scenarios.
- **Further Reading:** We will not cover the mathematical derivations of these methods in class, but you can find their details, applications, and statistical tables in Oehlert (2000). For instance, comparing all to treatments to the “best” treatment (section 5.5.2), not known in advance.

When you encounter these scenarios in your own research, modern statistical software (like the `emmeans` package in R) makes applying them straightforward. Refer to Oehlert (2000) for the underlying theory and to ensure your experimental design meets the assumptions for these specialized methods.

9 Testing Model Assumptions

Core Assumptions of the Errors

- The validity of ANOVA results, including contrasts and some forms of multiple testing, depends on our model assumptions.
- For the one-way ANOVA CRD model: $y_{ij} = \mu^* + \alpha_i + \epsilon_{ij}$.
- Our main assumptions for this model include:
 - All ϵ_{ij} are independent meaning that different observations in the same group are independent, and data from different groups are independent.
 - The variances of ϵ_{ij} 's are all the same.
 - The ϵ_{ij} 's are normally distributed.

We can combine these three assumptions into a single statement:

ϵ_{ij} 's are an independent and identically distributed (i.i.d.) random sample from $N(0, \sigma^2)$.

- Listing the assumptions in decreasing order of importance:
 1. **Independence** (most important). Achieved by randomization.
 2. **Constant variance** (next most important). Verified by checking residuals.
 3. **Normal distribution** (least important). Verified by checking residuals.

Probably all of these assumptions are never *exactly* true, we are more concerned with serious violations of these assumptions.

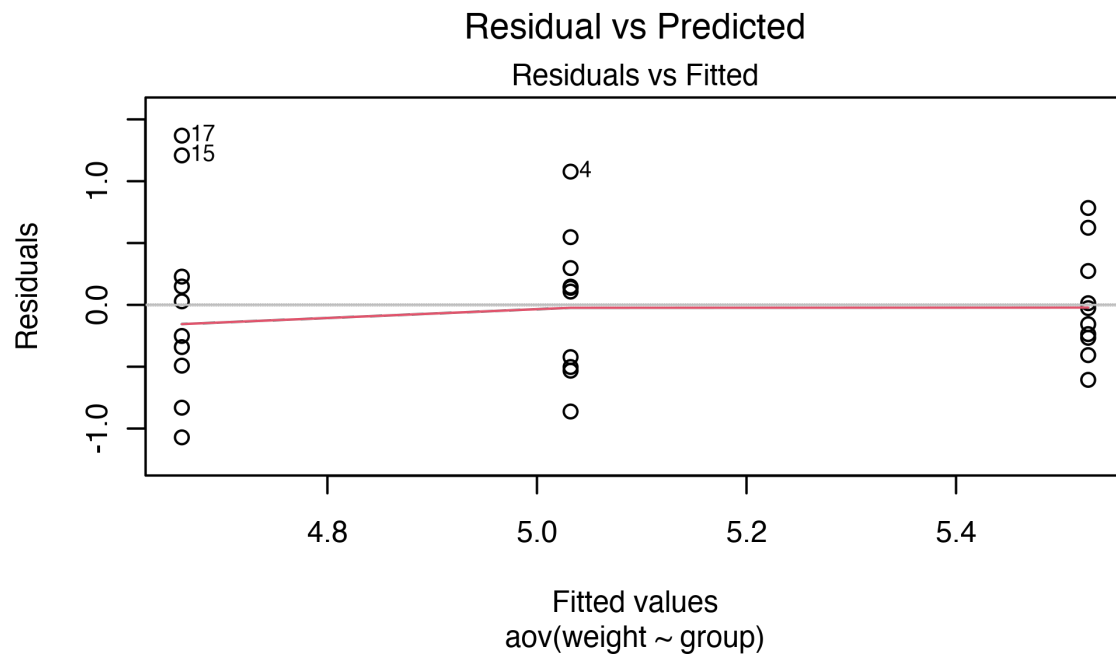
Diagnosing Violations

- Essentially all methods for diagnosing violation of assumptions are based on study of the *residuals*.
- Conceptually: $r_{ij} = \text{observed}_{ij} - \text{predicted}_{ij}$. Intuitively, if your model is appropriate, r_{ij} should resemble ϵ_{ij} .
- **Tests vs. Plots:** There are tests for normality and equal variances, but they have limitations (small sample sizes lack power; large sample sizes detect trivial problems).
- Therefore, formal tests are *supplemental*, while plots are *principal*.

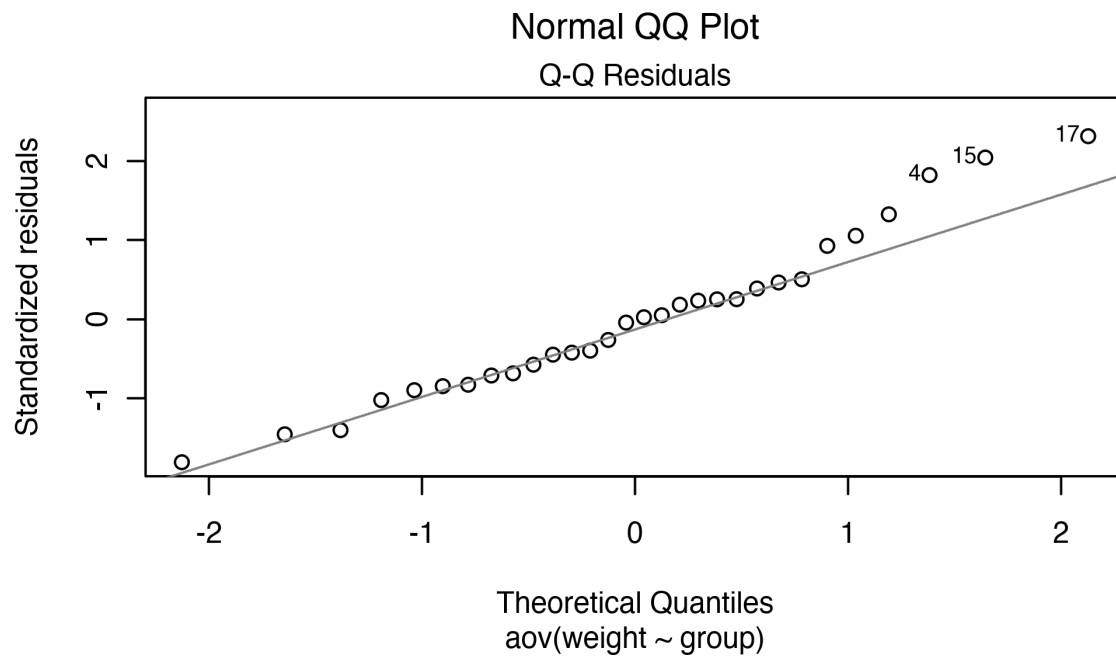
R Code: Principal Plots for Diagnostics

- We can generate the principal diagnostic plots (Residual vs Predicted and Normal QQ Plot) directly from our ANOVA model object in R.

```
# Using the pg_model (PlantGrowth) from earlier
# 1. Residual vs Predicted (Fitted) plot for constant variance
plot(pg_model, which = 1, main = "Residual vs Predicted")
```



```
# 2. Normal probability (QQ) plot for normality
plot(pg_model, which = 2, main = "Normal QQ Plot")
```



- **Supplemental Tests:** If you need formal statistical tests, you can use the Shapiro-Wilk test for normality and Levene's test for equal variances.

```

# Supplemental Test for Normality on residuals
# (Reject Normality assumption if small)
shapiro.test(resid(pg_model))

Shapiro-Wilk normality test

data:  resid(pg_model)
W = 0.96607, p-value = 0.4379

# Supplemental Test for Equal Variances (requires 'car' package)
library(car)
leveneTest(weight ~ group, data = PlantGrowth)

Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  1.1192 0.3412
      27

```

Checking Specific Assumptions

- **Independence:** Independence of the ϵ_{ij} 's cannot be checked by considering residuals.

Even when the ϵ_{ij} are independent, the residuals r_{ij} are *not*. Specifically, the r_{ij} are designed to sum to zero for each group $i \in 1 : g$:

$$\sum_{j=1}^{n_i} r_{ij} = 0 \quad \implies \quad r_{in_i} = - \sum_{j=1}^{n_i-1} r_{ij}$$

From this, you can actually show:

$$\text{Corr}(r_{i,j_1}, r_{i,j_2}) = -\frac{1}{n_i}, \quad \text{for all } j_1 \neq j_2.$$

The best way to protect against independence is proper experimental design and randomization. Randomization itself induces independence of the errors.

- **Variance and Normality:** Heteroskedasticity (changing variances) and non-normality usually come hand-in-hand. Both can be checked by formal tests (supplemental), or by plots (primary approach). Key plots include:
 - Normal QQ-plots
 - Residuals vs predicted plot
 - Residuals in time order plot

Remedies: Data Transformations

- The principle remedial tool is re-expression (transformation) of the response.
- If two transformations $\tilde{y}_1 = f_1(y)$ and $\tilde{y}_2 = f_2(y)$ have a linear relationship ($\tilde{y}_2 = \frac{\tilde{y}_1 - a}{b}$), they are completely equivalent for correcting assumptions.
- For log transformations, the transformations $\tilde{y}_1 = \log_e(y)$ and $\tilde{y}_2 = \log_{10}(y)$ are equivalent transformations because $\tilde{y}_1 = \ln(10)\tilde{y}_2$.
- Another common transformation includes the Box-Cox power family of transformations for a positive response variable is defined as:

$$\tilde{y} = \begin{cases} (y^p - 1)/p, & p \neq 0 \\ \ln(y), & p = 0 \end{cases}$$

Note that $(y^p - 1)/p$ is equivalent to just y^p .

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References

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